MH6003

Cryptography

Course information

1 — WEEK 1 — Start

7 — WEEK 7 — Mid-term

13 — WEEK 13 — Final exam

CONTACT

Your instructors

INSTRUCTOR

Prof. Wang Huaxiong

HXWang@ntu.edu.sg

INSTRUCTOR

Dr. Su Le

LSU1@e.ntu.edu.sg

3 academic units

Evening session: 6:30–10:00 pm

COURSE SCHEDULE

Weekly topics / weeks 1–6

Week	Topics / milestones
1	Basic concepts of cryptography and its applications
2	Symmetric-key cryptography: block ciphers and stream ciphers
3	Symmetric-key cryptography: block ciphers and stream ciphers
4	Cryptographic hash functions and message authentication codes
5	Cryptographic hash functions and message authentication codes
6	Public-key cryptography: encryption and signatures

Weekly topics / weeks 7–13

Week	Topics / milestones
7	Public-key cryptography: encryption and signatures; mid-term test
8	Public-key cryptography: encryption and signatures
9	Pseudorandom number generation
10	Identification schemes and entity authentication
11	Key distribution and key agreement schemes
12	Key distribution/agreement and public-key infrastructure (PKI)
13	Revision (about 1 hour), followed by the final exam (2 hours)

ASSESSMENT

How the course is graded

20%

Mid-term test

Individual

30%

Group project

Team

50%

Final examination

Individual

Total: 100% of the course grade

Exam arrangements

Mid-term test

WEEK 7

1 hour

CLOSED BOOK

Individual assessment

Final examination

WEEK 13

2 hours

CLOSED BOOK

Individual assessment

Week 13: approximately one hour of revision, followed by the two-hour final exam.

TEAMWORK

Group project / plan ahead

Teams of 3–4 students. The project contributes 30% of the course grade.

By week 7

Form your group
Choose your topic

→

Weeks 8–11

Four-week
project period

8 9 10 11

→

End of week 11

Submit
the project

State each member’s individual contribution.